Yipeng (Harry) Wang

Princeton University
Department of Mathematics

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Academic Interests

I am an instructor at Princeton University. My research interests lie in geometric analysis and various aspects of differential geometry. Currently, I am focused on investigating problems related to minimal surfaces and the geometry of manifolds with positive scalar curvature.

Education

Columbia University

- Ph.D. in Mathematics (2022–2026).
- Advisor: Simon Brendle.

Columbia University

- M.A. in Mathematics, 2024
- B.S. in Applied Mathematics, *Magna Cum Laude* (Class of 2022).

Publications and Preprints

1. **On a variant of the Penrose Inequality** (joint with Sven Hirsch). preprint.
2. **spacetime PMT** (joint with Simon Brendle). preprint.
3. **PMT** (joint with Simon Brendle). preprint.
4. **Area minimizing capillary cones** (joint with Benjy Firester, Raphael Tsiamis). preprint.
5. **stability capillary cone** (joint with Benjy Firester, Raphael Tsiamis). preprint.
6. **On fill-ins with scalar curvature bounded from below and an inequality of Hijazi-Montiel-Roldán** (joint with Simon Brendle, Raphael Tsiamis). Submitted.
7. **Uniqueness of Cylindrical Tangent Cones $\mathbf{C_{p,q} \times R}$** (joint with Benjy Firester, Raphael Tsiamis). journal reference.
8. **Rigidity Results Involving Stabilized Scalar Curvature**. to appear.
9. **Fill-Ins of Tori with Scalar Curvature Bounded from Below**. Submitted.
10. **Effective volume growth of three-manifolds with positive scalar curvature**. To appear in Proc. AMS.
11. **Scalar Curvature Rigidity of Polytopes**. Oberwolfach Report (2024).

[1]Updated August 9, 2026

12. **On Gromov's rigidity theorem for polytopes with acute angles** (joint with Simon Brendle). Journal für die reine und angewandte Mathematik (Crelles Journal), 2025. https://doi.org/10.1515/crelle-2025-0048.
13. **Scalar curvature rigidity of warped product metrics** (joint with Christian Bär, Bernard Hanke and Simon Brendle). SIGMA 20 (2024), special issue dedicated to Jean-Pierre Bourguignon on the occasion of his 75th birthday.

Invited Talks

1. *PMT.* Courant, GA, Septermber, 2026.
2. *PMT II.* Princeton, DGGA, Septermber, 2026.
3. *PMT.* Princeton, DGGA, Septermber, 2026.
4. *PMT.* Lehigh?, GA, Septermber, 2026.
5. *PMT.* Penn State, ICM satellite, July, 2026.
6. *PMT.* Online (Peking Westlake University), Geometric Analysis Seminar at BICMR, May/Jun ....
7. *PMT.* CUNY, Geometric Analysis Seminar, Apr/May. 2026.
8. *PMT.* Toronto Fields, Geometric Analysis Seminar, May. 2026.
9. *PMT.* Hamilton Mcmaster, Geometric Analysis Seminar, May. 2026.
10. *PMT.* Rutgers, Geometric Analysis Seminar, Apr. 2026.
11. *PMT.* Stonybrook, Geometric Analysis Seminar, Apr. 2026.
12. *Fill-in Estimate with Scalar Curvature bounded from Below.* Online (Peking University), Geometric Analysis Seminar at BICMR, Oct. 2025.
13. *Fill-In Problems with Scalar Curvature Constraints.* Online, Convergence of Scalar Curvature Workshop, Jul. 2025.
14. *Fill-In Problems with Scalar Curvature Constraints.* Universität Augsburg, Oberseminar Differentialgeometrie, May. 2025.
15. *Rigidity Results Involving Stabilized Scalar Curvature.* CUNY Graduate Center, Geometric Analysis Seminar, May. 2025.
16. *Rigidity Results Involving Stabilized Scalar Curvature.* Online, Peking-Westlake Geometric Analysis Seminar, Apr. 2025.
17. *Fill-In Problems with Scalar Curvature Constraints.* Stanford University, Geometry Seminar, Jan. 2025.
18. *Rigidity Results in Scalar Curvature.* Courant New York University, Geometric Analysis and Topology Seminar, Nov. 2024.
19. *Rigidity Results in Scalar Curvature.* Hong Kong University of Science and Technology, Geometry Seminar, Jun. 2024.
20. *Rigidity Results in Scalar Curvature.* ETH Zürich, CHANGE CHallenges in ANalysis and GEometry, Jun. 2024.
21. *Scalar Curvature Rigidity of Polytopes.* Oberwolfach Mathematics Institute, Analysis, Geometry and Topology of Positive Scalar Curvature, Feb. 2024.
22. *On Gromov's rigidity theorem for polytopes with acute angles.* Online (IHES), GNOSC Not Only Scalar Curvature Seminar, Oct. 2023.

Teaching Experience

Princeton University

- Instructor, Calculus (Fall 2026)

Columbia University

- Teaching Assistant, Modern Analysis I (Spring 2026)
- Teaching Assistant, Fourier Analysis (Fall 2025)
- Instructor, Calculus II (Summer 2025)
- Teaching Assistant, Ordinary Differential Equations (Spring 2025).
- Teaching Assistant, Ordinary Differential Equations (Fall 2024).
- Teaching Assistant, Partial Differential Equations (Spring 2024).
- Teaching Assistant, Fourier Analysis (Fall 2023).
- Teaching Assistant, Calculus IV (Spring 2022).
- Teaching Assistant, Fourier Analysis (Fall 2021).

Organization

Seminars

- Co-organizer of *Student Geometric Analysis Seminar: Lagrangian Mean Curvature Flow*, with Raphael Tsiamis (Fall 2025). Seminar website
- Co-organizer of *Student Geometric Analysis Seminar: Min-max theory for minimal submanifolds*, with Raphael Tsiamis (Spring 2025). Seminar website
- Co-organizer of *Student Geometric Analysis Seminar: Regularity Theory of the Monge-Ampère Equation*, with Raphael Tsiamis and Jingbo Wan (Spring 2024). Seminar website.
- Co-organizer of *Student Geometric Analysis Seminar: Mean Curvature Flow*, with Raphael Tsiamis and Jingbo Wan (Fall 2023). Seminar website.
- Co-organizer of *Student Geometric Analysis Seminar: Dihedral Rigidity*, with Aaron Chow and Jingbo Wan (Spring 2023). Seminar website.

Programming Skills

Mathematica, Python, C++